Upper Confidence Bound (UCB)

Reinforcement Learning Notes

1 The Multi-Armed Bandit Problem

We have d options ("arms"), e.g. d ad versions to show to users. At each round $n = 1, \dots, N$ we must choose one arm $i \in \{1, \dots, d\}$ to play. Playing arm i returns a reward $r_i(n) \in \{0, 1\}$ (e.g. whether the user clicked the ad).

Each arm i has an unknown, fixed expected reward μ_i (its true click-through rate). The goal is to maximize cumulative reward over N rounds, which requires balancing:

- **Exploitation** — keep choosing the arm that looks best so far.
- **Exploration** — try other arms to check whether they are actually better, since estimates from few trials are unreliable.

UCB resolves this exploration/exploitation trade-off deterministically, by choosing at each round the arm with the highest *upper confidence bound* on its expected reward.

2 Notation

At the start of round n, for each arm i define:

- $N_i(n)$: number of times arm i has been selected in the rounds before n.
- $R_i(n)$: sum of rewards obtained from arm i so far.
- $\bar{r}_i(n) = \dfrac{R_i(n)}{N_i(n)}$: empirical average reward of arm i (exploitation term).

3 Confidence Interval and Upper Bound

By Hoeffding's inequality, the true mean μ_i lies within

$$\bar{r}_i(n) - \Delta_i(n) \ \le\ \mu_i \ \le\ \bar{r}_i(n) + \Delta_i(n)$$

with high probability, where the confidence radius is

$$\Delta_i(n) = \sqrt{\frac{3}{2} \cdot \frac{\log(n)}{N_i(n)}}.$$

Note that $\Delta_i(n)$:

- **shrinks** as $N_i(n)$ grows — the more we've tried arm i, the more confident we are in $\bar{r}_i(n)$ (exploitation dominates).

- **grows slowly** with n (the round number) — arms we haven't tried in a while regain some uncertainty, encouraging exploration.

The *upper confidence bound* for arm i at round n is

$$\mathrm{UCB}_i(n) = \bar{r}_i(n) + \Delta_i(n).$$

This is an optimistic estimate of μ_i: the best-case value consistent with the data observed so far. This is the "optimism in the face of uncertainty" principle.

4 Algorithm

Algorithm 1 Upper Confidence Bound

$N_i \leftarrow 0$, $R_i \leftarrow 0$ for all $i = 1, \ldots, d$
for $n = 1, \ldots, N$ **do**
  for $i = 1, \ldots, d$ **do**
    if $N_i > 0$ **then**
      $\bar{r}_i \leftarrow R_i / N_i$
      $\Delta_i \leftarrow \sqrt{\frac{3}{2} \log(n) / N_i}$
      $\mathrm{UCB}_i \leftarrow \bar{r}_i + \Delta_i$
    else
      $\mathrm{UCB}_i \leftarrow +\infty$ ▷ force every arm to be tried once
    end if
  end for
  play arm $i^* = \arg\max_i \mathrm{UCB}_i$
  observe reward r; update $N_{i^*} \leftarrow N_{i^*} + 1$, $R_{i^*} \leftarrow R_{i^*} + r$
end for

The $+\infty$ initialization (`1e400` in the notebook code) guarantees every arm is sampled at least once during the first d rounds, since an untried arm always beats any finite UCB value.

5 Correspondence with the Implementation

The notebook (`upper_confidence_bound.ipynb`) implements exactly this loop over the `Ads_CTR_Optimisation.cs` dataset, with $N = 10000$ rounds and $d = 10$ ads:

Math	**Code**
$N_i(n)$	`numbers_of_selections[i]`
$R_i(n)$	`sums_of_rewards[i]`
$\bar{r}_i(n)$	`average_reward`
$\Delta_i(n)$	`delta_i`
$\mathrm{UCB}_i(n)$	`upper_bound`
$\arg\max_i \mathrm{UCB}_i(n)$	`ad` (tracked via `max_upper_bound`)

The final histogram of `ads_selected` shows that, as n grows, the algorithm converges to selecting the ad with the highest true CTR far more often than the others — the exploration term shrinks for the best arm as its estimate becomes reliable, while inferior arms are selected less and less.

6 Key Takeaways

- UCB is a *deterministic* strategy: no randomness is used to pick an arm, unlike Thompson Sampling.
- It requires updating estimates after every single round (cannot be batched/deferred), since $N_i(n)$ and n both change per round.
- It has strong theoretical regret bounds: cumulative regret grows only logarithmically in N.